

DTOF RGB-D CAMERA · S SERIES

ULTRA-WIDE MRDVS S11 RGB-D CAMERA

A 140° horizontal field of view in a 130-gram package. One S11 covers what two narrower sensors used to – from 0.1 metres out to 6, with ± 1 cm accuracy at 2 m.



■ WHY THIS CAMERA

- **140° × 56° depth field of view** – the widest in the S Series; one S11 replaces two narrower sensors
- **± 1 cm at 2 m, ± 2 cm at 6 m** – the tightest accuracy in the S Series, held consistently across the full range
- **0.1 m minimum range** – a small near-field blind zone for edge following and wall tracking
- **Direct time-of-flight depth** – valid range on every pixel across black, reflective and textureless surfaces
- **Ethernet, USB 3.0 or MIPI CSI-2** – the most flexible connectivity in the series (MIPI on volume orders)
- **Typ. <4 W, 130 g, 90 × 25 × 25 mm** – the smallest and lightest camera in the S Series
- **100 kLUX ambient light immunity** – unaffected by sunlight or floor-to-ceiling glazing
- **IP54, -20 °C to 60 °C**, Class 1 laser per IEC 60825-1, with a C / C++ / ROS 1 / ROS 2 SDK

140°

HORIZONTAL FOV

0.1–6 m

WORKING RANGE

± 1 cm

ACCURACY @ 2 M

130 g

WEIGHT

■ APPLICATION SCENARIOS



CLEANING ROBOTS

Edge following and boundary detection



QUADRUPED ROBOTS

All-round perception and terrain sensing



WHEELCHAIRS & CARTS

Low-speed outdoor collision prevention



AMR & AGV CHASSIS

Wide-angle chassis obstacle avoidance

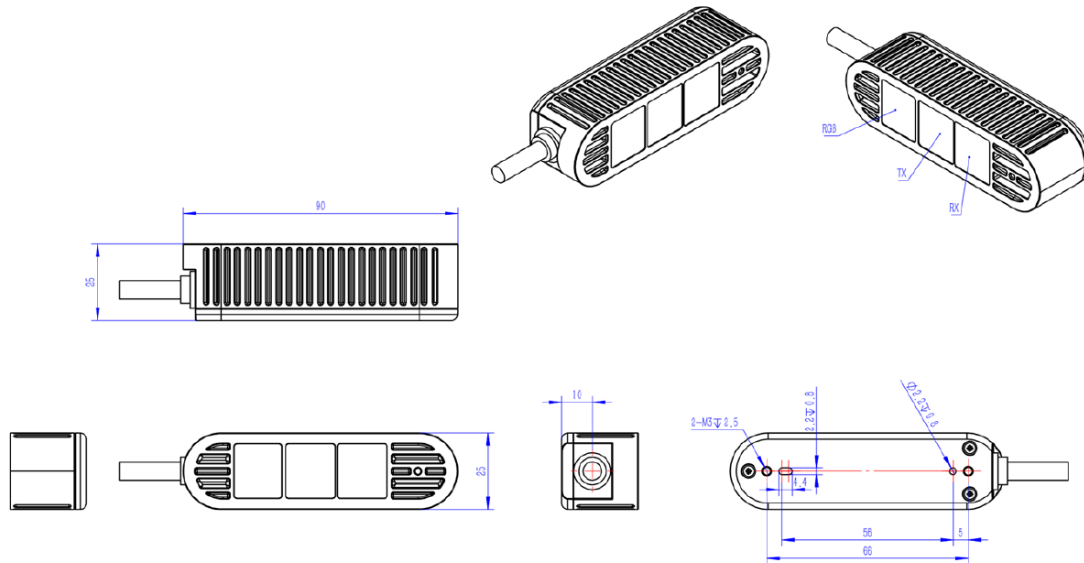
MRDVS S11 – SPECIFICATIONS



PARAMETER	TECHNICAL DATA
Sensing technology	dToF (direct time-of-flight) SPAD array
Laser wavelength	940 nm
Laser safety level	Class 1 (IEC 60825-1:2014)*
Output format	Depth / RGB / IR amplitude map / point cloud
Depth resolution & frame rate ^①	240 × 96 @ max. 15 fps (typ. 10 fps)
ToF FOV (H × V)	140° × 56°
RGB resolution & frame rate ^①	1280 × 1080 @ max. 15 fps (typ. 10 fps)
RGB FOV (H × V)	120° × 110°
Range	0.1–6 m (10–90% reflectivity, 100 kLUX)
Accuracy ^②	±1 cm @ 2 m; ±2 cm @ 6 m
Ambient light resistance	100 kLUX
IMU	—
Interface	Ethernet / USB 3.0

PARAMETER	TECHNICAL DATA
MIPI CSI-2 available	Yes (volume orders only)
Time synchronisation	PTP
Power supply	12–28 V DC or 5 V USB 3.0
Average power consumption	Typ. <4 W (max. 13 W)
Dimensions (L × W × H)	90 × 25 × 25 mm
Weight	130 g
IP rating	IP54
Operating temperature	–20 °C to 60 °C
Storage temperature	–40 °C to 70 °C
Humidity	Max. 90% RH, operating and storage
SDK support	C / C++ / ROS 1 / ROS 2
OS compatibility	Windows 7/8/10/11, Linux, Arm Linux
GUI software	LxCameraViewer

OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. CE, Class 1 laser safety (IEC 60825-1:2014), RoHS (2011/65/EU Annex II and amendment (EU) 2015/863) and REACH (EC No 1907/2006) certification is in progress, with completion planned for Q3 2026. * Marked values are design targets pending completion of certification. Contact MorpheusTEK to confirm current certification status.

Notes. ① Typical (typ.) frame rate is the recommended operating condition for optimal thermal and power performance; maximum (max.) frame rate is technically achievable but increases power draw and temperature rise and is not recommended for general use. Contact MorpheusTEK for application-specific evaluation. ② Accuracy may degrade with distance and depends on object reflectivity and surface roughness.

Installation. Mount the camera centred in its opening with no housing edge, rib or bracket intruding into the nominal field of view; FOV truncation causes fixed blind areas and false close-range readings. Keep the sensor at least 10 cm from highly reflective surfaces. Apply moderate, even mounting torque. Provide ventilation in enclosed spaces. Clean the optical window on a regular schedule in dusty or humid environments.

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